

AMENDMENT NO. 2 MAY 2013
TO
IS 10490 : 1983 SPECIFICATION FOR BREECH LOADING SHOT GUNS,
SINGLE AND DOUBLE BARREL

(Page 2, clause 3) — Substitute the following for the existing clause:

Sl No.	Name of Parts	Material
i)	Barrel	IS 5517 : 1993 ‘Steel for hardening and tempering’ Steel Designation 42Cr4Mo3 or 40C8 or 45C8 or 50C8. If made by forging then IS 1875 : 1992 ‘Carbon steel billets, blooms, slabs and bars for forging’, Grade 45C8 or 55C8. Heat treatment -To be heat treated to achieve mechanical properties given below: UTS=700 to 850 MPa 0.2 Percent proof stress = 480 MPa, <i>Min</i> IS:13015, C1S1R1 Inclusion rating- As per IS 4163 : 2004 Thick series : Nil Thin series : 1.0 (<i>Max</i>) Hardness — 210 to 270 HV
ii)	Body	IS 5517 : 1993 Grade 55C8 or 40Ni 14 or IS 3930 :1994 ‘Flame and induction hardening steels’ Grade 37 Mn 6 and Si shall be 0.10 to 0.35 percent.
iii)	Trigger	IS 5517 : 1978 Steel Designation 35Mn6Mo3 or 35Mn6Mo4 or 42Cr4Mo2 or 40Cr4 or IS 4368 : 1967 ‘Alloy steel billets, blooms and slabs for forgings for general engineering purposes’ Grade 40Cr 1 or 40Cr1Mo28. Heat treatment — To be heat treated to achieve mechanical properties given below: UTS = 800 MPa, <i>Min</i> 0.2 Percent proof stress = 600 MPa, <i>Min</i> Elongation on 5.65√A = 16 percent, <i>Min</i> Izod impact = 55 J Hardness = 250 HV generally and 400 HV locally at the catch point
iv)	Sear	IS 5517 : 1993 Steel Designation 31Ni10Cr3Mo6 or IS 3431 : 1982 ‘Steel for volute, helical and laminated springs for automotive suspension’ Grade 50Cr4V2 or 55Si7 or 60Si7 or 65Si7 of IS 3885 (Part I) : 1992 ‘Steel for manufacture of laminated springs (railway rolling stock): Part I Flat sections (<i>first revision</i>)’ Steel Designation 55Si7. Heat treatment — To be heat treated to achieve hardness of 350-450 HV generally and 600-700 HV locally at catch point.
v)	Hammer/ Pin Firing	IS 5517 : 1993 Steel Designation 55C8 or 40Cr4 or 37C 15. If made by forging of IS 1875 : 1992 Grade 55C8 or IS 4368:1967 Steel Designation 40Cr 1 or IS 2004 : 1991 Grade 55 C8. Heat treatment — To be heat treated to achieve hardness of 350-450 HV generally and 500-550 HV locally at striking face

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vi)	Main spring	IS 3431 : 1982 [*] Steel for manufacture of volute, helical and laminated springs for automotive suspension (<i>second revision</i>) [*] Steel Designation 55Si7 or 60Si7 or 65Si7 or IS 3885 (Part I) : 1982 [*] Steel for the manufacture of laminated springs (railway rolling stock): Part 1 Flat sections (<i>first revision</i>) Steel Designation 55Si7 Heat treatment — To be heat treated to achieve hardness 450-550 HV
vii)	Fastener fore-end	IS 7887 : 1992 Grade-8 with hardness restricted to 115 to 160 HV
viii)	Spring lever top	IS 3431 : 1982 Grade 55Si7 or 60Si7 or 65Si7 or IS 3885 (Part I) : 1992 Steel Designation 55Si7. Heat treatment — To be heat treated to achieve hardness of 500 -550 HV.
ix)	Catch hook	IS 5517 : 1993 Steel Designation 55C8. Heat treatment — To be heat treated to achieve hardness of 400-450 HV.
x)	Stock butt and stock fore-end	IS 7549 : 1975 [*] Timber half wrought for sporting rifles [*]

(Page 3, clause 4.5) — Substitute the following for the existing clause:

‘4.5 Mechanical Safety

Details are given below:

- a) Ensure that the shot gun is unloaded and it is ‘Safe’. Following action needs to be taken:
 - 1) Safety lever should be pushed for withdrawal of safety; and
 - 2) Hold the weapon by left hand at the fore end and put thumb on the locking lever. Locking lever should be rotated anticlockwise and the action is opened. Look through the barrel/barrels and ensure that the chamber is empty.
- b) Ensure that the acceptance ‘Proof’ mark as indicated below are stamped/engraved on the locations as indicated below:
 - 1) Body
 - aa) Proof-Mark
SBBL : Ashoka Lions
DBBL : Ashoka Lions
 - 2) Barrel/Barrels
 - aa) Proof-Mark
SBBL : Ashoka Lions
DBBL : Letter ‘P’ surmounted by an Ashoka Chakra and
Ashoka Lions in both the barrels
- c) Registered number of shot guns should tale in each of its components where the registered number exists. Normally in Shot Gun the registered number exists in the following components:
 - 1) Body
 - 2) Barrel/Barrels
 - 3) Fastener fore end

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4.6 Maintenance

Details are given below:

- a) Before Firing Maintenance — When it is intended to fire a shot gun, the following action should be taken before firing:
 - i) Ensure that the gun is not loaded after opening the breech;
 - ii) With a piece of cloth wipe out oil from external parts;
 - iii) Using a flannelette to clear the bore dry and inspect the bore;
 - iv) Check the functioning of the trigger and firing pins with open breech firing pin should not be sticking or jam, it should be withdrawn while clearing the breech; and
 - v) Close the breech and check functioning of safety lever.
- b) After Firing Maintenance :
 - 1) Barrel/Barrels
 - i) Clean externally with a dry cloth, remove oil;
 - ii) Run hot water through the barrel/barrels for 20 min;
 - iii) With the help of a cleaning rod and flannelette clean the bore dry till mirror/bright finish;
 - iv) Remove residue from recesses using a toothbrush;
 - v) When perfectly dry, apply lightly gun oil internally using the rods and flannelette; and
 - vi) Apply oil lightly on external surface.
 - 2) Action Body : Since the action body is fitted with butt, care must be taken to avoid application of oil/ grease on the butt.
 - i) Clean the metallic parts with a dry cloth. It is not necessary to dismantle it;
 - ii) Remove oil from slots using a tooth brush or find piece of cloth;
 - iii) Operate the mechanism by checking the trigger movements, movement of hammer spring and functioning of firing pin;
 - iv) Check firing pins for bend or damage and or jam and replace if required;
 - v) With the help of a flannelette cloth apply oil lightly to all metallic parts; and
 - vi) Clean an oil metallic parts of fore end fastener.
 - 3) Butt and Fastener:
 - i) Clean with dry cloth; and
 - ii) Apply licensed oil with a piece of cloth and allow it to dry for 4 to 6 h.’

(Page 4, clause 8, lines 2 and 4) — Substitute ‘IS 6005 : 1998’ for ‘IS 6005 : 1970’.